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Runoff and soil erosion from highway construction spoil deposits: A rainfall simulation study

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ABSTRACT

To measure soil loss from these road construction spoils, 30 rainfall simulations involving five rainfall intensities were conducted to study runoff and soil erosion processes on a road in China. Twenty-five of the rainfall simulations were carried out on loose materials five different slopes. Another five rainfall simulations were conducted on compacted road materials with a slope of 30.6%. The results show that soil bulk density has a positive effect on runoff rate, and the effect of slope on runoff rate changes with rainfall intensity. Higher soil bulk density increases soil loss under low rainfall intensity, but decreases soil loss when rainfall intensity is high. Soil loss is positively related to slope, but the relationship differs from that in prior studied of croplands and disturbed lands. Equations for predicting soil erosion from loose materials and compacted road are developed.

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1. Introduction

With economic development, the need for road construction generally increases. However, road construction also has detrimental effects on local environments, with soil erosion often a major issue. Most studies on erosion have concentrated on unpaved forest road network. Relatively fewer studies have been conducted on the soil erosion problems during highway construction, and these have mainly on embanks (Persyn et al., 2004; Xu et al., 2009) and construction sites (Kalainesan et al., 2009). Generally, when road building is taking place there are for periods construction sites, unpaved road surface, cut areas, fill areas, embankments and construction spoil deposits (excess materials from excavation). The soil loss from road construction is generally presumed to equal the aggregated loss from these, but there are few studies of this. Dong et al. (2011) finds, however, that nearly 2–5 million m³ of construction spoils are produced per 100 km of highway construction in China.

In another context, where have been studies of soil erosion processes from construction spoils, namely from slag at mines, these have shown that factors affecting erosion are not only different from those of croplands or natural hillslopes, but also vary significantly across sites. Schroeder (1987), for example, in studying the effects of slope gradient on erosion from two mine slag surfaces concluded that the slope function used in the universal soil loss equation (USLE) underestimated the effect for slopes under 9%, and overestimated for steeper slopes. Further, McIsaac et al. (1987) found that USLE slope steepness overpredicts the observed erosion for mines where the slopes are greater than 9%, and the relationship between soil loss and slope steepness varies considerably across the experimental sites. Riley (1995) examining the Ranger Uranium Mine in Australia, concluded that the unmanaged surface of the dump is 10–100 times more erodible than adjacent natural hillslopes, compared to Schroeder's estimate of the erodibility of mines of approximately 81% of that calculated from the USLE.

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Discrepancies between estimates of erosion from slag at mines and from excess material excavated for road construction continues to hamper the ability to adequately assess erosion from road construction. Both process-based landscape evolution models such as SIBERIA and empirical models such as RUSLE are used to predict farmland and mine waste soil erosion, nevertheless, modifying them to assess erosion from road construction requires a reassessment of their parameters (Zhang et al., 2004, 2008). As a consequence, it is necessary to determine the erosion affect factors from highway construction spoils to calibrate modeled soil loss from road construction areas. Because runoff and erosion data are rarely collected under natural rainfall, rainfall simulation is employed in this study.

2. Materials and method

The experiments were conducted near the town of Fuxian, 114 km south of the city of Yanan (Fig. 1), within Shanxi Province of China. The area is semiarid, with annual rainfall 569 mm, and with most of the precipitation occurring between June and September. Its elevation is 1120 m, and the area is used as a spoil deposit for excess excavated material from road construction from the eighth segment of the Qing-Lan highway. It varies in gradient from 19% and 78% and is covered with silt loam. The spoil deposit is made up of one central spoil pile, covering more than 1.3 hectare with about $8 \times 10^4 \text{ m}^3$ construction spoils. Two treatments of surface condition examined: one comprises loose materials (LM) with steep slopes that have been dumped by shovel loaders; the other comprises soil with greater bulk density and less steep slopes that is compacted by vehicles to serve as construction roads (CR). The construction spoils had been dumped for about two years, and most of the study area is devoid of vegetation.

Mayer style rainfall simulators are used that are similar to those described in Mayer and Harmon (1979), but modified for a larger scale rainfall simulation and for rapid movement between research sites. The trough comprises the main body of the rainfall simulator and is 32 cm in width and 25 cm in depth. The length of the trough is determined by the number of nozzles, of which there were three for this study, spaced 1.1 m apart. During the rainfall simulation, nozzles could swing along the length of the trough in order to match rainfall and raindrop size. Water used tests was from a refillable storage container located at the end of the trough that was pushed through a 5 cm diameter PVC hose by a 370 W pump. During each test, the trough was elevated 2.5 m above ground level and the water pressure of each nozzle controlled at 0.5 MP. The rainfall simulators were spaced 1.5 m apart. Each test area was bounded by iron plates connected to form rectangular plots. To prevent leakage, the iron plates were inserted to a depth of at least 10 cm. At the base of the plots, a V-shaped iron trough was installed to permit runoff sampling.

The simulated rainfall tests were carried out on six plots (Table 1). Plots a through e were set up on dumped loose materials (LM), and plot f was on a construction road (CR). Five rainfall intensities were selected: 0.6, 1.22, 1.74, 2.33 and 2.79 mm min^{-1} , for a total of 30 rainfall simulation events that were carried out (five events times six plots). Runoff rate reached a steady state 20–30 min after runoff started. The first sample was collected as soon as the initial runoff was generated with samples subsequently collected at two-minute intervals. The volume of samples varied from 1 L to 10 L, and the sediment concentration in runoff was determined gravimetrically after oven-drying at 105°C . If the sample volume was greater than 1 L, the sample was stirred for one minute, and a 1 L subsample taken to determine the sediment concentration.

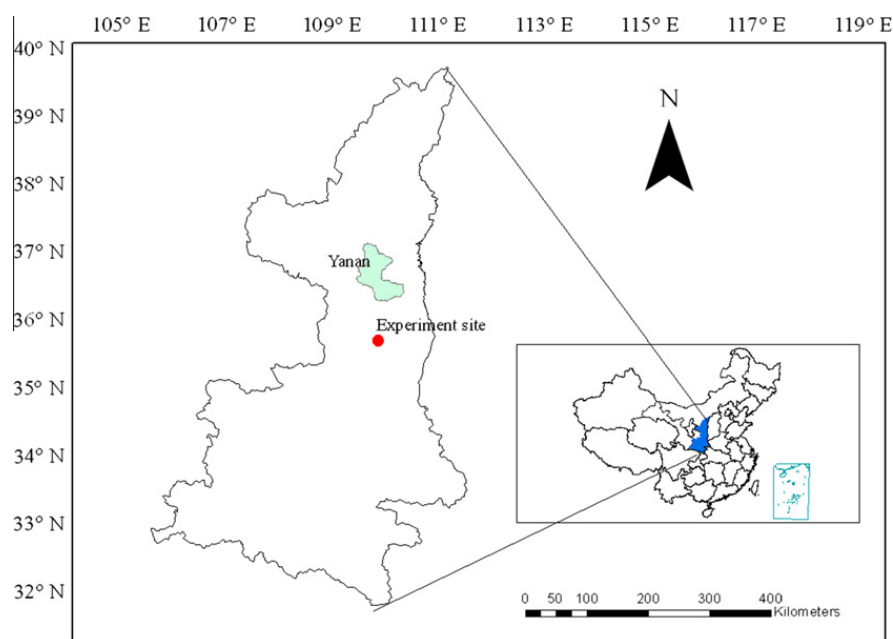


Fig. 1. Location of experimental site in Shan Xi province on a loess plateau.

Table 1

Soil properties and other attributes of the six plots.

	Plot a	Plot b	Plot c	Plot d	Plot e	Plot f
Plot width (m)	2.1	2.1	2.1	2.1	2.1	2.1
Plot length (m)	10.0	10.0	10.0	4.2	3.8	10.0
Soil bulk density (g cm ⁻³)	1.37	1.22	1.27	1.41	1.24	1.66
Slope (%)	19.4	30.6	46.6	57.7	78.1	30.6
sand (%)	18.4	19.7	17.4	20.4	16.9	17.2
silt (%)	65.2	62.7	69.1	64.4	68.5	70.1
clay (%)	15.9	16.6	12.5	14.7	13.6	12.0

Sand: 0.05–2.0 mm, silt: 0.002–0.05 mm and clay: <0.002 mm.

The soil loss rate was calculated from the runoff discharge and sediment concentration. Prior to each rainfall simulation, the incised rills developed from the previous rainfall test were carefully refilled and smoothed to minimize the impact of the previous simulation. To provide similar initial soil moisture for each rainfall test, all the rainfall tests are conducted for a 'wet soil run' condition: prior to each rainfall test, a simulation with rainfall intensity 0.6 mm min⁻¹ was employed, and this prior-simulation was terminated as soon as runoff started. Simulated rainfall tests were initiated 12 h after this prior-simulation.

For each experimental plot, the bulk density was determined from soil in the 0–5 cm layer within a 100 cm³ soil cutting ring. The initial soil moisture and soil moisture after the rainfall test in the 0–5 cm layer of each plot was determined by oven-drying the soil at 105 °C.

To isolate the effect of soil bulk density and slope on runoff and soil loss rates, partial correlation coefficients were applied in this study:

$$r_{xy,z} = \frac{r_{xy} - r_{xz}r_{yz}}{\sqrt{1 - r_{xz}^2}\sqrt{1 - r_{yz}^2}} \quad (1)$$

where $r_{xy,z}$ is the partial correlation coefficient between x and y holding z constant; r_{xy} is the correlation coefficient of x and y ; r_{xz} is the correlation coefficient of x and z ; r_{yz} is the correlation coefficient of y and z ; x , y and z are parameters. Rainfall erosivity (R) in MJ mm ha⁻¹ h⁻¹ is selected to represent the impact of rainfall on soil erosion. The R factor is the El_{30} for each rainfall test, and it is calculated with the guideline for RUSLE (Renard et al., 1997):

$$e_m = 0.29[1 - 0.72 \exp(-0.05i_m)] \quad (2.1)$$

$$E = \sum e_r P_r \quad (2.2)$$

$$R = \sum El_{30} \quad (2.3)$$

where e_m is kinetic energy in MJ ha⁻¹ mm⁻¹ of rain, i_m is intensity in mm h⁻¹, P_r is rainfall volume in mm, I_{30} is the maximum rainfall intensity in a 30 min period during the event, in mm h⁻¹. In order to eliminate the impact of varying slope length, the slope length of each plot was standardized to 10 m with the equation proposed by Wischmeier and Smith (1978):

$$L = \left(\frac{\lambda}{22.13} \right)^{0.5} \quad (3)$$

where θ is the slope angle and λ is the slope horizontal length in meters. Determination coefficient (r^2) and Model efficiency (M_E) were selected to assess the capability of the model in predicting the soil loss rate, calculated as (Nash and Sutcliffe, 1970):

$$M_E = 1 - \frac{\sum (y_i - y_{ci})^2}{\sum (y_i - \bar{y}_i)^2} \quad (4)$$

where M_E is model efficiency, y_i is the i th observed value, y_{ci} is the i th calculated value.

3. Results

3.1. Runoff rate

Experimental results showed that the runoff per unit area was controlled by rainfall intensity on all tested plots. Mean runoff rate increased in response to increasing rainfall intensity (I). The relationship could be described very well using the equations (Fig. 2). All coefficients of determination indicate that more than 98% of the variance in mean runoff rate could be explained by rainfall intensity.

Similar to rainfall intensity (I), soil bulk density also had a positive effect on mean runoff rate, with the partial correlation coefficients declining with increased rainfall intensity. When the rainfall intensity was greater than 1.74 mm min⁻¹, the

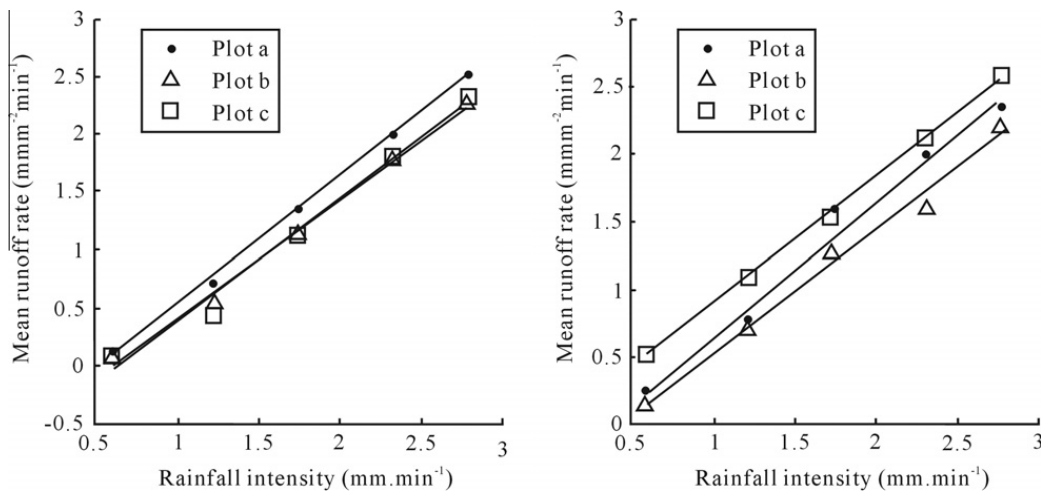


Fig. 2. Relationship between rainfall intensity and mean runoff rate.

effect of soil bulk density on mean runoff rate was not statistically significant at 2.33 and 2.79 mm/min rainfall tests. While the effect of slope on mean runoff rate was more complex, it too varied with rainfall intensity. This may be due to surface conditions, such as sealing and rilling, changing with slope under different rainfall intensities (Fox et al., 1997). No partial correlation coefficient of slope and rainfall intensity was significant. This indicates that soil bulk density had a greater effect on mean runoff rate than slope. Both the effects of soil bulk density and slope on runoff rate were masked by higher rainfall intensity.

3.2. Soil loss rate

For a given rainfall intensity, soil loss rates were similar between plots (Fig. 3). Two types of soil loss processes were observed. In plot d, when the rainfall intensity was lower than 1.22 mm min⁻¹, the soil loss rate increased slowly with time until it gradually reached a steady state (type 1). When the rainfall intensity was greater than 1.74 mm min⁻¹, there was a sharp rise in soil loss rate shortly after the onset of runoff. Generally, the loss fell sharply after this initial peak, and thereafter showed little change or slowly declined over time (type 2). This is because under high rainfall intensities, the runoff rate can reach its peak in a short time and with sufficient transport capacity to flush the more erodible materials away from the surface. After this, the soil loss rate declined.

Besides rainfall intensity, other factors that induce type 2 soil loss include slope and soil bulk density. Plot b, c and e have similar soil bulk density, but under the 1.74 mm min⁻¹ rainfall test, a type 2 process took place in plot e while type 1 occurred in plots b and c. Although plot b had a steeper slope than plot a, a type 2 process was observed in plot a while plot b showed a type 1 process under the 1.74 mm min⁻¹ rainfall test (Fig. 3a and b). Thus, both steeper slope and larger soil bulk density may both induce a type 2 process.

Soil bulk density had a positive effect on the soil loss rate when rainfall intensity was lower than 1.74 mm min⁻¹. When rainfall intensity was greater than 2.33 mm min⁻¹, the soil loss rate declined with increasing soil bulk density (Table 2). Fullen (1985) concluded that erosion rates would increase on compacted soils, and that compacted soils would lower the threshold rainfall intensity at which rainfall became erosive. However, greater dry bulk density not only increases the runoff rate, but also reduces the soil detachability and transportability (Adekalu et al., 2006). As a result, soil loss rates may not always decrease or increase with declining soil bulk density (Ekwue and Harrilal, 2010).

3.3. Effects of slope on soil loss rate

To minimize the effects of soil bulk density, soil loss data for rainfall intensities of 1.74 and 2.33 mm min⁻¹ (i.e., the highest *P* values in Table 2) were selected to examine the relationship between slope and soil loss rate. Soil loss data under the two rainfall intensities were divided by soil loss data for plot a to be normalized to a 19%; close to that used in former studies. The average of the normalized soil loss data under the two rainfall intensities was used to fit the slope steepness factor:

$$S = 2.302 \sin \theta + 0.639 \quad (5)$$

where *S* is the normalized soil loss, and θ is the slope of each plot. To compare the slope steepness factor from croplands (*S*_{Li}) and disturbed lands (*S*_{Mc}), equations proposed by Liu et al. (1994) and Mclsaac et al. (1987) were also normalized to 19% slope. The results are shown in Fig. 4 and show that *S*_{Li} overestimates the *S* factor for the construction spoils studied here, especially for steep slopes. Although Mclsaac et al. (1987) concluded that the slope effect on soil erosion from disturbed

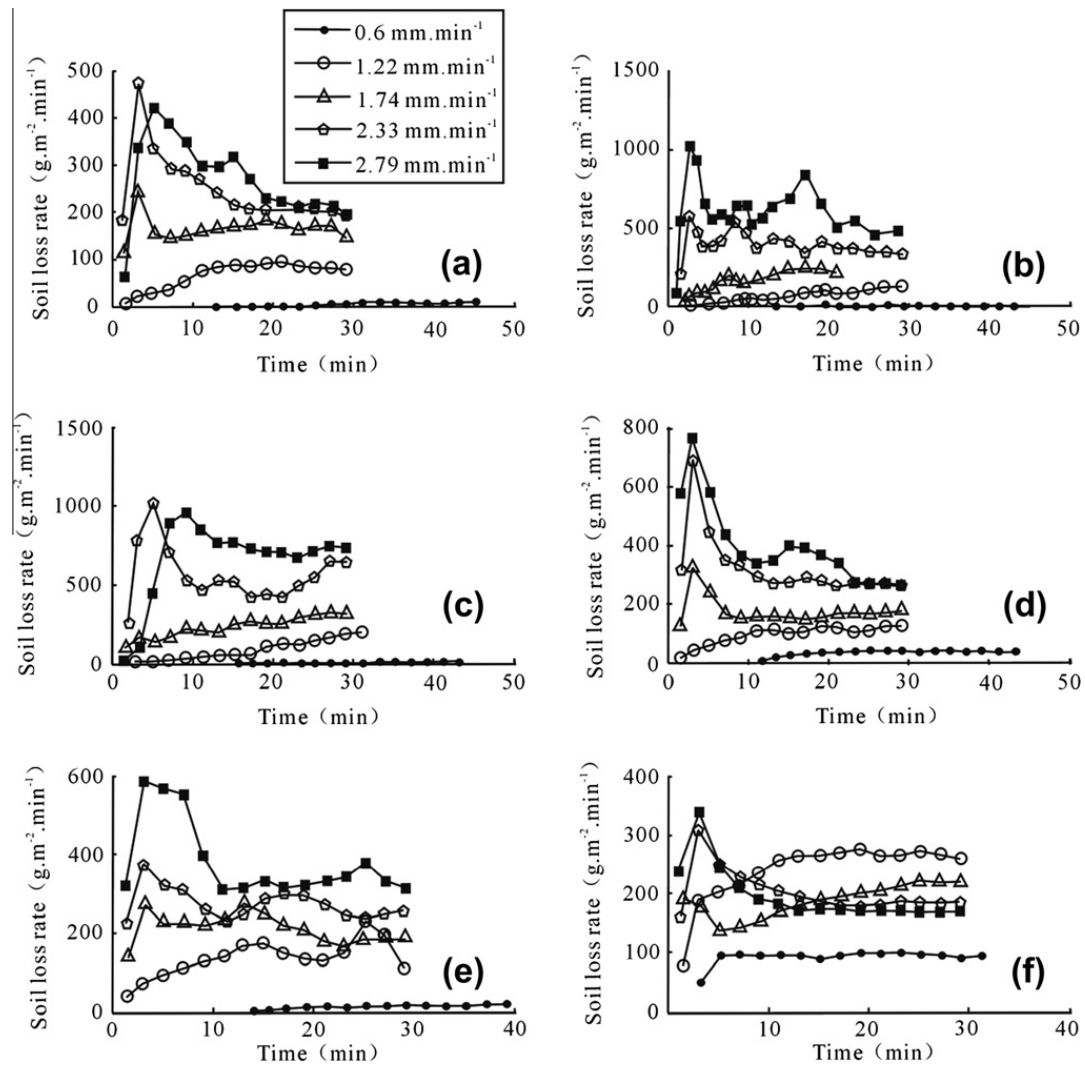


Fig. 3. Soil loss rate as a function of time for different plots.

Table 2

Partial correlation coefficients of soil bulk density versus mean soil loss rate under varying rainfall intensity.

I (mm min ⁻¹)	0.6	1.22	1.74	2.33	2.79
Mean soil loss rate (P_c)	0.968**	0.811*	0.407	-0.530	-0.784
p	0.006	0.047	0.472	0.304	0.105

* $p < 0.05$; ** $p < 0.01$: slope length standardized to 10 m.

areas varies widely across different experimental sites, the trend of the slope effect on soil loss encountered here is even lower than the lower boundary proposed by Mclsaac et al. (1987).

To account for the effects of soil bulk density on soil erosion, two equations are used to predict soil loss under rainfall intensities lower or greater than 1.74 mm min⁻¹:

$$A_{LM1} = \frac{SL}{1000} (0.86\rho + 3.42)EI_{30} \quad I \leq 1.74 \text{ mm min}^{-1} \quad (6.1)$$

$$A_{LM2} = \frac{SL}{1000} (6.39 - 1.81\rho)EI_{30} \quad I > 1.74 \text{ mm min}^{-1} \quad (6.2)$$

where A_{LM1} and A_{LM2} are calculated soil loss in kg m⁻², and ρ is soil bulk density in g cm⁻³.

Because the characteristics of CR are different from LM, predicting soil erosion on CR with Eq. (6) (CA_{LM}) may underestimate soil loss under low rainfall intensities, while overpredicting soil loss under high rainfall intensities. However, CA_{LM}

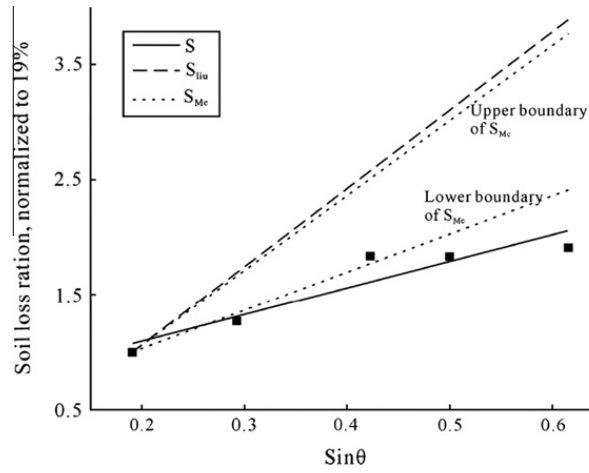


Fig. 4. Slope steepness factor, normalized to 19%, for highway construction spoils, cropland and disturbed land.

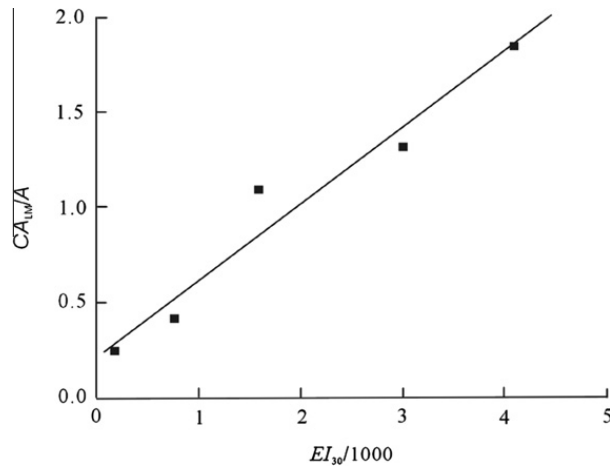


Fig. 5. Comparison between calculated soil loss rate and observed soil loss rate.

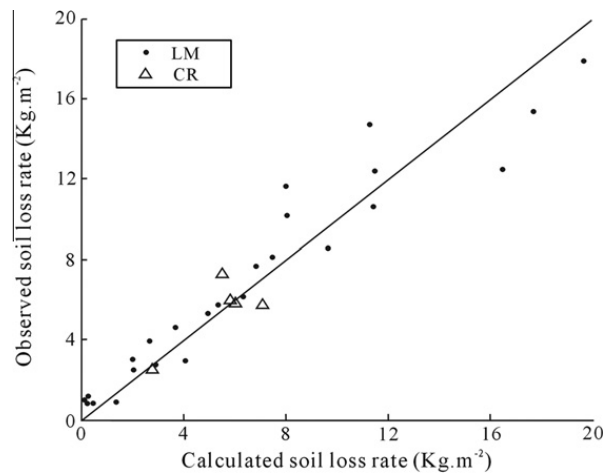


Fig. 6. Regression of EI_{30} and CA_{LM}/A .

divided by the observed soil erosion data (A) is observed to fit with EI_{30} very well (Fig. 5). Thus, the fitted equation may be regarded as a revised factor F :

$$F = 4.1 \times 10^{-4} EI_{30} + 0.21 \quad (7)$$

Furthermore, an equation for predicting soil erosion from CR could be expressed as:

$$A_{CR} = A_{LM}/F \quad (8)$$

where A_{CR} is the calculated CR soil loss rate (kg m^{-2}).

The equations predict soil erosion from excess materials of roadway construction fairly well (Fig. 6). The modeled efficiency of soil erosion estimates for LM and CR are 0.91 and 0.51, respectively. Owing to the complex CR surface conditions, the soil erosion prediction for CR is less successful than that for LM.

4. Conclusion

This study has looked at the erosion processes from construction spoils and the overall erosion caused by highway construction. Both rainfall intensity and soil bulk density were positively related to runoff, while the effect of slope on runoff varied with rainfall intensity. Rainfall intensity explained more than 98% of the variation of runoff, and soil bulk density had a greater effect on runoff than slope. The relationship between soil loss rate and slope can be expressed as a linear equation, and this relationship is different than those previously described for croplands and mines.

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References

- Adekalu, K.O., Okunade, D.A., Osunbitan, J.A., 2006. Compaction and mulching effects on soil loss and runoff from two southwestern Nigeria agricultural soils. *Geoderma* 137, 226–230.
- Dong, J.Z., Zhang, K.L., Zhang, W.X., Shao, S.G., 2011. Highway waste dump shape and regional differences. *Bulletin of Soil and Water conservation (in Chinese)* 3 (2), 163–167.
- Ekwue, E.I., Harrilal, A., 2010. Effect of soil type, peat, slope, compaction effort and their interactions on infiltration, runoff and raindrop erosion of some Trinidadian soils. *Biosystems Engineering* 105, 112–118.
- Fox, D.M., Bryan, R.B., Price, A.G., 1997. The influence of slope angle on final infiltration rate for interrill conditions. *Geoderma* 80, 181–194.
- Fullen, M.A., 1985. Compaction, hydrological processes and soil erosion on loamy sands in east Shropshire, England. *Soil and Tillage Research* 6, 17–29.
- Kalainesan, S., Neufeld, R.D., Quimpo, R., Yodnane, P., 2009. Sedimentation basin performance at highway construction sites. *Journal of Environmental Management* 90, 838–849.
- Liu, B.Y., Nearing, M.A., Risse, L.M., 1994. Slope gradient effects on soil loss for steep slopes. *Transactions of the ASAE* 37, 1835–1840.
- Mayer, L.D., Harmon, W.C., 1979. Multiple-intensity rainfall simulator for erosion research on row sideslopes. *Transactions of the ASAE* 22, 100–103.
- Mclsaac, G.F., Mitchell, J.K., Hirschi, M.C., 1987. Slope steepness effects on soil loss from disturbed lands. *Transactions of the ASAE* 30, 1005–1012.
- Nash, J.E., Sutcliffe, J.V., 1970. River flow forecasting through conceptual models part I – A discussion of principles. *Journal of Hydrology* 10, 282–290.
- Persyn, R.A., Glanville, T.D., Richard, T.L., Laflen, J.M., Dixon, P.M., 2004. Environmental effects of applying composted organics to new highway embankments: Part 1. Interrill runoff and erosion. *Transactions of the ASAE* 47, 463–469.
- Renard, K.G., Foster, G.A., Weesies, D.A., McCool, D.K., Yoder, D.C., 1997. Predicting soil erosion by water: a guide to conservation planning with the revised universal soil loss equation (RUSLE). *Agriculture Handbook NO. 703*. USDA, Washington, DC.
- Riley, S.J., 1995. Aspects of the differences in the erodibility of the waste rock dump and natural surfaces, Ranger Uranium Mine, Northern Territory, Australia. *Applied Geography* 15, 309–323.
- Schroeder, S.A., 1987. Slope gradient effect on erosion of reshaped spoil. *Soil Science Society of America Journal* 51, 405–409.
- Wischmeier, W.H., Smith, D.D., 1978. Predicting rainfall erosion losses – a guide to conservation planning. *Agricultural Handbook*. 537. USDA, US Government Printing Office, Washington, DC.
- Xu, X.L., Liu, W., Kong, Y.P., Zhang, K.L., Yu, B.F., Chen, J.D., 2009. Runoff and water erosion on road side-slopes: effects of rainfall characteristics and slope length. *Transportation Research Part D – Transport and Environment* 14, 497–501.
- Zhang, K.L., Peng, W.Y., Yu, B., 2004. Erodibility of agricultural soils on the Loess Plateau of China. *Soil and Tillage Research* 76, 157–165.
- Zhang, K.L., Shu, A.P., Xu, X.L., Yang, Q.K., Yu, B., 2008. Soil erodibility and its estimation for agricultural soils in China. *Journal of Arid Environments* 72, 1002–1011.